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Barriers to sustainable rural electrification: A PESTEL-based qualitative case study of Mae Hong Son Province, Thailand

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ABSTRACT

Universal access to affordable and reliable electricity remains a cornerstone of Sustainable Development Goal 7, yet many remote regions continue to face persistent barriers to electrification. Mae Hong Son Province in northwestern Thailand represents the nation's lowest electrified region, with 22.2% of its main villages remaining unserved. This study applies the PESTEL analytical framework—Political, Economic, Social, Technological, Environmental, and Legal—to systematically examine the multidimensional barriers to achieving full electrification in the province. Data was collected through document analysis and semi-structured interviews with government agencies, technical experts, and residents. A total of 39 barriers were identified across the six dimensions. While several factors, such as inadequate funding and grid inaccessibility, are consistent with challenges in other developing contexts, Mae Hong Son presents distinct barriers related to protected forest regulations and the legal status of Hill Tribe settlements. Institutional obstacles—particularly lengthy and complex permitting procedures—emerged as the most critical constraint. The study provides the first comprehensive, PESTEL-based assessment of rural electrification barriers in Thailand and delivers policy-relevant insights to support inclusive, sustainable, and context-specific electrification strategies that strengthen energy equity and socio-economic resilience in remote communities.

Introduction

Located in the far northwest corner of Thailand, Mae Hong Son (MHS) Province shares a 483-kilometer border with the Republic of the Union of Myanmar. Covering an area of approximately 12,780 km^2 (Mae Hong Son Province, 2025a), it ranks as Thailand's eighth largest province by size but its 287,644 residents (National Statistics Office (Thailand), 2025a) ranks 69th of the country's 76 provinces in population. The province's terrain is predominantly mountainous, with elevations ranging from 100 to 2000 m above sea level (Mae Hong Son Province, 2025a). Over 85% of its land area is covered by natural forests, making MHS a leading eco-tourism destination for both domestic and international visitors (Tourism Authority of Thailand, 2025).

Despite its scenic landscapes and rich cultural heritage, MHS remains one of Thailand's poorest provinces, with 24.6% of its population living below the national poverty line—the highest rate in the country (United Nations Development Programme, 2024). Poverty and energy deprivation are closely linked, and MHS has Thailand's lowest electrification rate. As of April 2024, only 77.8% of the province's main villages had access to electricity, leaving 22.2% of villages unelectrified. The household electrification rate stands at 86%, meaning roughly one in seven households still lack access to electricity (Mae Hong Son Province Energy Office, Ministry of Energy, 2024). Fig. 1 illustrates the village electrification rates across the province's seven districts.

Although the Thai government has made repeated attempts to electrify these remote villages, a significant number remain non-electrified. As described by United Nations (UN) Sustainable Development Goal (SDG) 7, providing access to electricity would vastly improve the lives of these villagers. Energy access also represents a key enabler toward achieving other SDGs including No Poverty (SDG 1), Zero Hunger (SDG 2), Good Health and Well-Being (SDG 3), Quality Education (SDG 4), Clean Water and Sanitation (SDG 6), Reduced Inequalities (SDG 10), and Climate Action (SDG 13) (United Nations Division for Sustainable Development Goals, Department of Economic and Social Affairs, 2021).

Despite the many non-electrified villages in MHS Province, both the International Energy Agency (IEA) and the World Bank's Energy Sector Management Assistance Program (ESMAP) report that Thailand achieved 100% access to electricity as of 2023 (International Energy Agency

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Mae Hong Son Province - 781 Total Villages*			
415 Main Villages*		366 Subvillages	
323 electrified (77.8%)	92 non-electrified (22.2%)	154 electrified (42.1%)	212 non-electrified (57.9%)

Pang Mapha District - 55 Total Villages			
38 Main Villages		17 Subvillages	
22 electrified (57.9%)	16 non-electrified (42.1%)	2 electrified (11.8%)	15 non-electrified (88.2%)

Pai District - 82 Total Villages			
62 Main Villages		20 Subvillages	
56 electrified (90.3%)	6 non-electrified (9.7%)	8 electrified (40.0%)	12 non-electrified (60.0%)

Mueang Mae Hong Son District - 127 Total Villages*			
68 Main Villages*		53 Subvillages	
60 electrified (88.2%)	8 non-electrified (11.8%)	33 electrified (62.3%)	20 non-electrified (37.7%)

Khun Yuam District - 77 Total Villages			
43 Main Villages		34 Subvillages	
41 electrified (95.3%)	2 non-electrified (4.7%)	18 electrified (52.9%)	16 non-electrified (47.1%)

Mae La Noi District - 127 Total Villages			
69 Main Villages		58 Subvillages	
59 electrified (85.5%)	10 non-electrified (14.5%)	21 electrified (36.2%)	37 non-electrified (63.8%)

Mae Sariang District - 190 Total Villages			
77 Main Villages		113 Subvillages	
52 electrified (67.5%)	25 non-electrified (32.5%)	53 electrified (46.9%)	60 non-electrified (53.1%)

Sop Moei District - 129 Total Villages			
58 Main Villages		71 Subvillages	
33 electrified (56.9%)	25 non-electrified (43.1%)	19 electrified (26.8%)	52 non-electrified (73.2%)

*Does not include Mae Hong Son Municipality which is 100% electrified

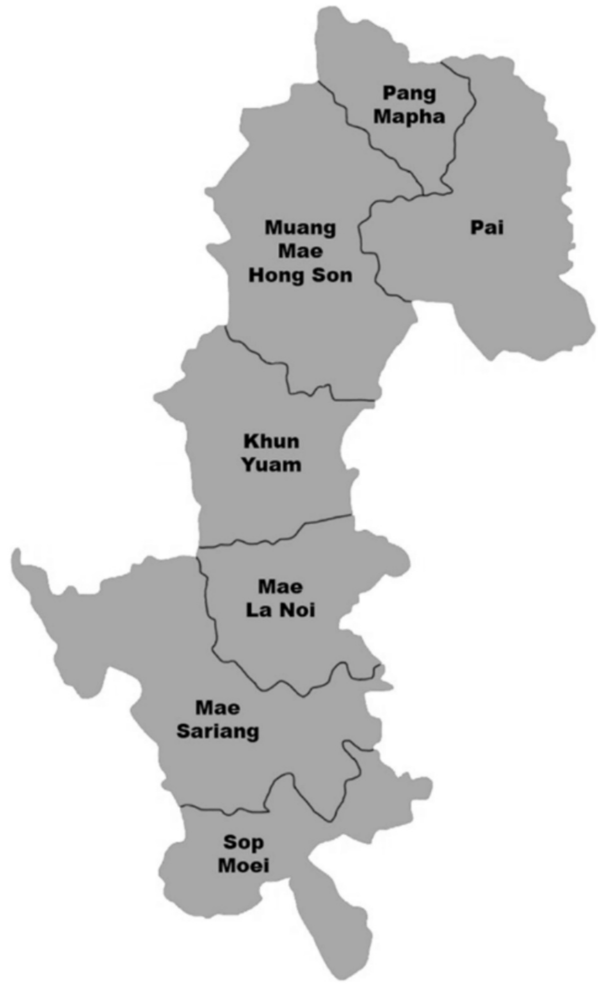

Fig. 1. MHS Province village electrification by district.
Adapted from Mae Hong Son Province Energy Office, Ministry of Energy (2024).

(IEA), 2025; World Bank, 2025). This can be attributed to the relatively small total population of MHS's villages, making the percentage of the population nationwide without electricity statistically zero. This paper explores the reasons why in a nation of almost complete electrification do so many residents of this one unique province still lack electricity. One reason is that traditional grid extension has proven difficult and financially unsustainable due to the province's rugged topography, dispersed settlements, and low population density. Consequently, recent initiatives have shifted toward decentralized renewable energy systems, including mini-grids and standalone solar home systems, which offer more practical and cost-effective solutions for remote areas.

The Thai government is expected to approve an update to the National Energy Plan (NEP) in early 2026, which includes an updated Power Development Plan (PDP) and Alternative Energy Development Plan (AEDP). The current draft PDP sets a goal to increase the share of renewables to 51% of total electricity generation by 2037 and 74% by 2050, a significant increase over the 15% total for 2023 (Asawinpongphan et al., 2025; United Nations Development Programme, 2025). The draft AEDP stipulates incentives to expand renewable energy production, including feed-in tariffs, tax exemptions, and grants for clean energy projects. These major policy changes should serve to encourage the expansion of both private and government/

Table 1
Literature summary.

Location	Year	Methodology	Barrier focus areas	Highlight
Worldwide	2021	Literature review, PESTEL (Zebra et al., 2021)	- Capital cost - Lack of effective laws - Weak government involvement - Financial/credit	PESTEL methodology and identification of common barriers in developing world
		Literature review, SALSA (Search, Appraisal, Synthesis, Analysis) (Streimikiene et al., 2021)	- Economic - Social - Institutional - Regulatory - Behavioral/psychological	Detailed, worldwide barrier list
Southeast Asia	2025	Literature review (Mamat et al., 2025)	- Upfront costs - Limited technical expertise - Policy inconsistencies - Socio-cultural resistance	Identifies barriers common in Southeast Asia
16 developing countries	2023	Literature review (Shyu, 2023)	- Financial - Technical - Project design and implementation	Identifies barriers focused on solar home systems
India	2025	MCDM PAPKIKA (Dash et al., 2025)	- Technical - Economic - Environmental - Social	Identifies most suitable rural wind energy options using MCDM
Bangladesh	2020	non-parametric test and literature review (Khan et al., 2020)	- Economic/financing - Technical - Customer service	Identifies barriers focused on solar home systems
Ghana	2025	Fieldwork - interviews and surveys (Arthur et al., 2025)	- Installation costs - Lack of financing - Maintenance challenges	Identifies barriers focused on solar PV installations
Philippines	2021	Literature review, fieldwork (interviews, site visits, community observations) (Quirapas-Franco & Taeihagh, 2021) QUIRAPAS-30	- Community-based barriers	Focus on small rural communities
Sub-Saharan Africa/Zambia	2025	Literature review (Makai & Popoola, 2025)	- Technical complexities - Economic and financial - Operations and maintenance - Policy and regulatory	Detailed study with numerous identified barriers
MHS Province	2007	Fieldwork, documentation review (Greacen & Martin, 2007)	- Technical/environmental - Social/economic - Policy/legal framework - Organizational	Identified barriers to Solar, Wind, Biomass, Hydro in MHS in 2007

community-based renewable energy projects (United Nations Development Programme, 2025), which are key to expanding electrification in MHS Province.

A defining feature of MHS Province is its large ethnic diversity, particularly its Hill Tribe populations who constitute over half of the province's residents. Approximately 16% of MHS's inhabitants also lack Thai citizenship, with the majority residing in Hill Tribe villages (Mae Hong Son Provincial Social Development and Human Security Office, Thailand, 2020). Notably, nearly all non-electrified villages are inhabited by Hill Tribe communities who in most cases do not possess formal ownership or land title. According to Thailand's Ministry of Energy (MOE) data from April 2024, all non-electrified main villages and 211 of 212 non-electrified sub-villages are located within protected areas, such as National Reserve Forests or National Parks (Mae Hong Son Province Energy Office, Ministry of Energy, 2024). The stringent regulatory framework governing development within these zones has created a complex administrative environment where obtaining project approval is both time-consuming and uncertain. This institutional constraint has emerged as a critical barrier to achieving full electrification in the province.

Table 1 provides a sampling of the extensive body of research examining barriers to rural electrification in developing countries, including technological, financial, and institutional dimensions. However, few studies have addressed cases where environmental protection policies and land tenure insecurity intersect to constrain electrification efforts. MHS Province represents a unique case study where rural energy poverty is not only a function of remoteness or cost, but also of legal and policy complexities associated with conservation and land rights. This presents both an analytical and policy gap in the literature, particularly within the Southeast Asian context.

This study seeks to address these gaps by systematically identifying and analyzing the barriers preventing MHS Province from achieving full electrification. Using the PESTEL framework (Political, Economic, Social, Technological, Environmental, and Legal dimensions), the study analyzes data from official documents and interviews with key government agencies, subject matter experts, and residents to achieve its research goals:

- Advance the existing literature by categorizing and interpreting the barriers to sustainable rural electrification that make MHS unique from Thailand's other provinces.
- Provide a comprehensive understanding of the province's unique electrification challenges to inform future policy and implementation strategies.
- Contribute to the broader discourse on sustainable rural electrification by highlighting how socio-political and environmental governance factors interact to shape access to modern energy.

The findings offer insights relevant not only to Thailand but also to other regions facing similar trade-offs between rural development and environmental conservation.

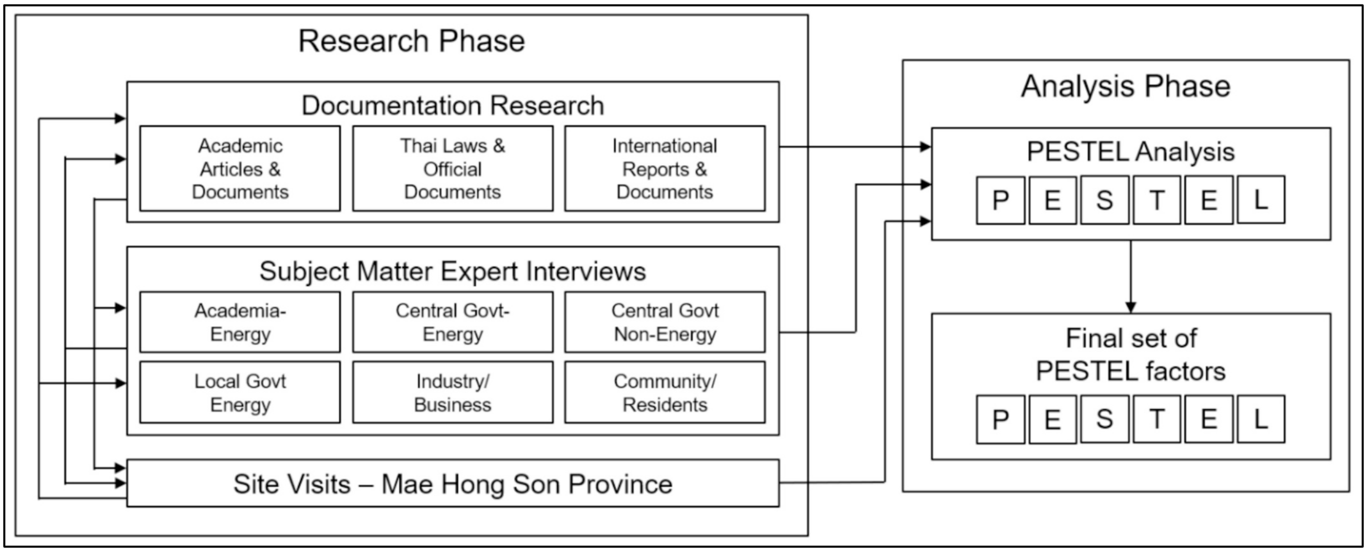

Fig. 2. Research methodology.

Material and methods

Literature review on sustainable rural electrification

The United Nations Development Programme (UNDP) *Sustainable Development Goals Report for 2025* listed 17 goals, with number seven stating "ensure access to affordable, reliable, sustainable, and modern energy for all" (United Nations Department of Economic and Social Affairs, 2025). 2025's *Tracking SDG 7-The Energy Progress Report* describes how global progress has been made since the initial report in 2018, but with progress toward 2030 targets remaining off track (IEA et al., 2025). Numerous studies conclude that a lack of access to electricity significantly constrains human development and well-being, representing a fundamental barrier to progress for a substantial proportion of the world's population. Without access to reliable and affordable energy, the lives of the energy poor are negatively impacted on a wide range of development indicators, including health, education, food security, gender equality, livelihoods, and poverty reduction (Bryce, 2023; Gashaye et al., 2025; Min et al., 2024; Shyu, 2020). Other studies have shown the correlation between introducing electrification and an increase in household income and a corresponding improvement in the quality of life in developing regions around the world, including Southeast Asia (Shyu, 2020), South Asia (Amin et al., 2022), the Philippines (Bertheau, 2020a), Ethiopia (Getie, 2020), Tanzania (Matimbwa & Mng'ong'o, 2024), and Morocco (Oulakhmis et al., 2025). However, Cravioto and other researchers have concluded that although many quality-of-life indicators improved for newly electrified villagers, economic progress did not always occur (Cravioto et al., 2020).

Greacen and Martin researched barriers to electrification in MHS Province in 2007, providing a baseline and starting point for this current research (Greacen & Martin, 2007). Although many factors have changed over the past 18 years, many 2007 barriers are still valid, and the changes themselves are of interest for further research. Many other researchers have analyzed barriers to rural electrification in other regions, which in some cases proved helpful in identifying MHS Province's electrification barriers. Zebra et al. employed the PESTEL analytical framework to examine the barriers to renewable energy system deployments in developing countries. Zebra identified the main barriers as the lack of specific legislation for mini-grids, weak involvement of local communities and private entities in decision making, lack of credit line facilities, and insufficient financial mechanisms (Zebra et al., 2021). Mamat's review identified barriers to rural electrification in Southeast Asia as upfront costs, limited technical expertise, policy inconsistencies, and socio-cultural resistance, while also identifying potential solutions (Mamat et al., 2025). Shyu (2023) detailed the lessons learned from World Bank Solar Home System (SHS) installations across 16 developing countries from 2000 to 2020, which led to the identification of barriers to the SHS project's success. Shyu focused on four categories of barriers: financial, technical, project design, and implementation (Shyu, 2023). Khan, T. et al. examined barriers to renewable energy projects in Bangladesh, also focusing on SHSs (Khan et al., 2020). Streimikiene et al. cited the shift toward decentralized energy production using renewable technologies in rural areas worldwide, citing a long list of barriers due to economic, social, institutional, regulatory, behavioral, and psychological factors while noting the rise of energy cooperatives in rural areas, as well as 'smart villages' (Streimikiene et al., 2021). Quirapas-Franco described how social aspects, namely community involvement, were instrumental in the success or failure of renewable energy projects in four non-electrified communities in the Philippines (Quirapas-Franco & Taeihagh, 2021). Several studies focus on rural electrification barriers in sub-Saharan Africa, which present concepts similar to those found in MHS Province (Arthur et al., 2025; Makai & Popoola, 2025).

Commonly cited barriers to expanding electricity access through traditional grid expansion include the situation where many non-electrified villages have small populations and are in remote areas with difficult access. Mini-grid and stand-alone off-grid solar solutions continue to be key to electrifying these areas due to the ease of deployment and the ability to cost-effectively meet the lower and more dispersed demand (IEA et al., 2025; Mosetlhe et al., 2025). Additional recent research indicates how stand-alone solar photovoltaic (PV) has emerged as the most viable option for electricity production in isolated rural areas. This includes Mamat's review, which concluded that solar PV systems represent the most viable off-grid solution for rural electrification in Southeast Asia after evaluating the potential and barriers of solar, wind, biomass, and hydropower systems (Mamat et al., 2025). Sustainable wind energy sources should also be considered as potential renewable energy production solutions. Dash et al. described a process using Multiple Criteria Decision Making (MCDM) techniques which considered technical, economic, environmental, and social factors to evaluate different wind power solutions (Dash et al., 2025).

Several studies have shown that hybrid renewable systems that include solar PV and other renewable energy technologies (RET) can be even more effective for isolated communities, as they overcome some of the shortcomings of solar and wind systems, such as intermittent production. Hybrid systems can also lower costs and decrease the size of

Table 2
Methodology details.

Research activity	Number/range	Years	Focus/examples
Documentation research			
Academic articles	36 cited articles	1982–2025	Energy/electrification, RETs, barriers
Thai laws	5 laws	1941–2019	Thailand national laws
Thai Govt. documents	13 agencies, 21 documents	2009–2025	MHS Province PAO, PEA, MOE, RFD, NPD, National Statistics Office
International documents	3 agencies, 6 documents	2009–2025	UN, UNDP, IEA, IRENA, ESMAP
Subject matter expert interviews			
Academia	7 experts, 10 interviews	2023–2025	Thailand-specific rural electrification.
Thai Govt.-Energy	8 experts, 13 interviews	2023–2025	PEA, MOE, DEDE, EGAT. Focus
Thai Govt.-Non-energy	4 experts, 4 interviews	2023–2025	RFD, NFD, Ministry of Tourism
Local government	14 experts, 18 interviews	2023–2024	Province, District, Subdistrict, Village
Industry/business	6 experts, 6 interviews	2023–2025	National/local RET companies
Community/residents	10 experts, 21 interviews	2023–2025	Electrified & non-electrified in MHS
Site visits			
Locations visited	7 subdistricts, 20+ villages	2018–2025	Mueang MHS, Pai Districts

system components (Channi et al., 2024; Zebra et al., 2021). Several studies describe the success of hybrid renewable systems featuring solar PV and other technologies in case studies on Racha Island, Thailand (Udayachalerm et al., 2024), isolated Philippine islands (Bertheau, 2020b), and in Indonesia (Tee et al., 2025).

These studies represent a portion of the extensive body of research literature examining the subject of rural electrification in developing regions worldwide. There are many common barriers shared by the world's non-electrified rural villages, such as a lack of funding, remote locations, and small populations. This study builds on the body of current research by focusing on the unique barriers and characteristics in MHS Province and how they combine to impede the goal of complete electrification.

Methodology

Fig. 2 illustrates the study's major components in its two phases. During the research phase, potential barriers to rural electrification in MHS Province were identified. The information gained during the research phase was then input into the analysis phase where electrification barriers were identified and categorized into one of the six PESTEL categories. The research was iterative, as information was gained it was used as feedback to identify further sources of information and continuously refine the PESTEL barrier list.

Research phase: fieldwork and data collection

The study's research phase employed a mixed qualitative approach to gathering information consisting of (1) a systematic documentary review, (2) semi-structured qualitative interviews with subject-matter experts and community members, and (3) direct field observations in selected villages. Fieldwork activities were conducted between December 2023 and May 2025. Documentary sources covered the period 1982–2025.

Semi-structured interviews were conducted with representatives from academia, national and provincial energy agencies, forestry and conservation departments, local governments, renewable energy providers, and residents or leaders from electrified and non-electrified villages. In total, 57 interviews were carried out with 33 experts and community representatives, as summarized in Table 2. Examples of interview sessions include discussions held with energy officials and local authorities on 12 December 2023 and 18 December 2024. Interviews followed a flexible guide aligned with the PESTEL framework to ensure consistent thematic coverage while allowing new issues to emerge.

A qualitative interviewing process was employed as it allowed for a deeper understanding of the issues concerning the wide range of barriers associated with rural electrification. The process was dynamic and flexible, allowing the interviewer to delve deeper into subject areas that may not have been anticipated at the start of the interview. The oral responses also allowed some participants to speak more freely about sensitive subjects such as lack of citizenship or land ownership. The interviewees also represented a broad segment of society, from highly educated experts in academia to experienced engineers and officials from government agencies to the largely uneducated rural village leaders and farmers living in the non-electrified areas. A rigid questionnaire would not be a fit for this broad spectrum of participants.

Field visits were conducted in seven subdistricts covering more than 20 villages, including several non-electrified villages located within protected forest zones. Observations focused on terrain conditions, settlement layout, accessibility, environmental constraints, and existing electrical infrastructure. These field observations were used to triangulate interview and documentary evidence.

Document analysis included legal frameworks, provincial and district statistical reports, MOE electrification datasets (up to 29 April 2024), watershed and forest protection zoning maps, and relevant project reports from 2009 to 2025. These sources provided the structural context for identifying policy and regulatory barriers. Table 2 provides further research phase details.

Analysis phase: PESTEL framework

The research phase identified the rural electrification barriers which serve as inputs to the analysis phase, which employed the PESTEL methodology. PESTEL is a mnemonic that, in its expanded form, denotes P for Political, E for Economic, S for Social, T for Technological, E for Environmental, and L for Legal. PESTEL provides an ideal analysis framework as it organizes the many barriers into logical categories using an established analytical process (Makos, 2023). The research was iterative, as the information gained was used as feedback to identify further sources of information and continuously refine the PESTEL barrier list.

Researchers commonly use PESTEL analysis for studies on electrification barriers as it provides a well-known framework to facilitate the identification, classification, and categorization of barriers. The resulting barriers are presented in categorized groups which facilitate multi-perspective and cross-sectoral analysis of the subject of interest, resulting in later identification of solutions and strategies to overcome the barriers.

The PESTEL analysis phase ran in parallel with the research phase. As factors related to electrification barriers were identified through research, they were categorized into one of the six PESTEL factor categories. The output from the PESTEL Analysis Phase was the final list of PESTEL factors, where each factor was named and numbered, and a written description was developed as described below. These barriers are presented in the following section.

- Political barriers include factors imposed by national, provincial, and local government agencies through policies and processes, and are denoted with the prefix “P”.
- Economic barriers include financial factors, including the economic situation, project funding, and system costs, and are denoted with the prefix “Ec”.
- Social barriers include factors from the social environment, such as cultural trends, demographics, and the population's attitudes toward potential solutions, and are denoted with the prefix “S”.

- Technological barriers include technical factors that impact rural electrification progress and are denoted with the prefix “T”.
- Environmental barriers include factors such as the unique geography and environment of MHS Province, as well as environmental policies, and the population's environmental concerns, and are denoted with the prefix “En”.
- Legal barriers describe laws and regulations that impact rural electrification progress and are denoted with the prefix “L”.

Barriers that are unique to a particular energy technology are further denoted as follows:

- Solar PV: “(S)”
- Wind: “(W)”
- Hydro: “(H)”
- Bioenergy: “(B)”.

Ethical considerations

All procedures involving human participants adhered to institutional and national guidelines for social research ethics. Prior to each interview, participants received a clear explanation of the study objectives, voluntary nature of participation, and the handling of personal information. Informed consent was obtained verbally from each participant. Interviewees were offered the option of anonymity; those who requested it have been anonymized in the manuscript and in all data records.

No personal, identifying, or sensitive information was collected beyond professional role, sector, or community position. Interview notes and audio recordings were stored securely on password-protected devices accessible only to the research team. The study complied with ethical guidelines of Naresuan University, and institutional confirmation of exempt status or supporting documentation can be provided to the journal upon request.

Participant recruitment

Participants were recruited using purposive sampling to ensure representation across stakeholders relevant to rural electrification. Experts were approached directly through their agencies, including central and local government organizations and agencies, renewable energy providers, and academic institutions. For community interviews, introductions were facilitated through local administrative officers and the Provincial Administrative Organization (PAO). Snowball sampling was used where appropriate to reach additional knowledgeable informants, such as village leaders and technicians.

Bias mitigation strategies

Several measures were implemented to minimize selection and interpretation bias:

1. Triangulation: Findings were cross-validated using three evidence streams—documentary sources, expert interviews, and field observations.
2. Diverse stakeholder sampling: Interviews covered actors across government, industry, academia, conservation agencies, and communities, reducing over-reliance on any single perspective.
3. Iterative coding: Two researchers independently coded interview notes using the PESTEL framework depicted in Fig. 2. Coding inconsistencies were resolved through discussion until consensus was reached.

Results

This section presents the results of the PESTEL analysis, which identified the barriers to rural electrification in MHS Province. These barriers were determined through a comprehensive analysis of documentation research, subject matter expert interviews, and information collected during site visits to MHS Province.

Political barriers to rural electrification

Four current political barriers to MHS Province's rural electrification were identified.

P1: government agencies involved with electricity are spread across multiple ministries and departments

Thailand has several electricity-related agencies that influence rural electrification. Most significantly for MHS Province, the Provincial Electricity Authority (PEA) falls under the Ministry of the Interior, while the Electricity Generating Authority of Thailand (EGAT) and Department of Alternative Energy Development and Efficiency (DEDE) fall under the MOE. Additionally, the two key agencies responsible for controlling access to non-electrified village lands are the Royal Forest Department (RFD) and the Department of National Parks, Wildlife and Plant Conservation (DNP), both under the Ministry of Natural Resources and Environment.

Although these organizations claim good cooperation at the local level (Chaipruek, 2023; Chansopa, 2023; Wailom, 2023), issues still arise due to the division and overlap of authority and responsibility between them. One subject matter expert explained:

> “*The electrification projects [from different agencies] are more separate than in conflict.*”
>
> MOE Plans and Policy Analyst, Expert Level (Tungpokanon, 2023)

> “*There are no conflicts between PEA and EGAT or the Forest Department in MHS, the process is slow, but it works.*”
>
> Director, MHS PEA Office (Chansopa, 2023)

One current example is the separate efforts at establishing a smart grid in MHS Province. Despite electricity distribution being a core PEA function, EGAT opened the Smart Grid Development Pilot Project in Pha Bong in the Mueang Mae Hong Son District in 2023 with the ability to monitor the grid, but not to control grid functions, as control falls under PEA's purview. PEA also operates a smart grid pilot project in Mae Sariang District. Future plans involve merging the various smart grid projects into a single smart grid for all of Mae Hong Son Province (Chansopa, 2023), a project which will coordination among the various agencies.

P2: differing priorities and political differences between the province governor and the PAO president and council

There can be political coordination issues at the provincial government level. As in all Thai provinces, the MHS Province governor and deputy governors are appointed by the Ministry of Interior and are not from the MHS area, and often not even from northern Thailand. However, the MHS PAO President and PAO Council are local residents elected by the province's citizens. Differing priorities and provincial as well as national politics can cause issues (Kamnuansilpa, 2025).

P3: differing priorities from Local Administrative Organizations (LAOs) at the subdistrict level

The Provincial Administrative Organization Act empowered subdistrict-level LAOs, which are led by a subdistrict headman who is an elected resident of the subdistrict (Provincial Administrative Organisation (PAO), 2003). The LAOs present their funding requests at the provincial level, but village electrification projects are not always the top priority (Prayadsap, 2024), as a local government officer explained:

> “*The first priority is to improve the roads from dirt to concrete. Second priority is to get clean water and electricity.*”
>
> Mueang MHS Deputy District Chief (Panchoo, 2023)

P4: lack of continuity for appointed government agency officials

Appointed government officials frequently rotate, especially in senior positions. Senior leaders such as the governor and deputy governors

Table 3
Total installed cost by technology, 2010 and 2024.

	Year 2010		Year 2024		Percent change
	USD/kW	THB/kW	USD/kW	THB/kW	
Bioenergy	3082	101,706	3242	106,986	5%
Hydropower	1494	49,302	2267	74,811	52%
Solar PV	5283	174,339	691	22,803	−87%
Onshore wind	2324	76,692	1041	34,353	−55%

Source IRENA (International Renewable Energy Agency (IRENA), 2025).

Table 4
Levelized cost of electricity by technology, 2010 and 2024.

	Year 2010		Year 2024		Percent change
	USD/kW	THB/kW	USD/kW	THB/kW	
Bioenergy	0.086	2.838	0.087	2.871	1.2%
Hydropower	0.044	1.452	0.057	1.881	29.5%
Solar PV	0.417	13.761	0.043	1.419	−89.7%
Onshore wind	0.113	3.729	0.034	1.122	−69.9%

Source IRENA (International Renewable Energy Agency (IRENA), 2025).

are from outside MHS and come for a short tour of duty, typically lasting from several months to a maximum of 3 years. This lack of continuity can impact progress as initiatives started by a predecessor may not be continued. For this reason, the constant rotation can serve as a barrier to progress (Mae Hong Son Provincial Energy Office, Thailand, 2023; Ninudomsak, 2023).

Economic barriers to rural electrification

Seven current economic barriers to MHS Province's rural electrification were identified:

Ec1: Lack of funding for electrification projects. Limited funding for electrifying the remaining non-electrified villages is an obvious barrier. If unlimited funding were available, electricity could quickly be provided to all villages, but this isn't realistic, given that funding is always limited and electrification competes with other priorities. While 2027 was stated as MOE's goal for 100% electrification of the main villages, more realistically, it will take at least five to ten more years past 2027 due to budget constraints and the other barriers (Wailom, 2023).
Ec2: Total installed cost (TIC) of RETs. As shown in Table 3, the International Renewable Energy Agency (IRENA) (2025) published TIC data for the four RETs relevant to MHS Province from 2010 to 2024. TIC includes all costs associated with designing, procuring, fabricating, and installing a project. The cost to extend the main grid via traditional distribution lines is discussed in the Environmental (En) section. TIC for each of the four RETs relevant to MHS Province is discussed individually below. Battery storage costs, which are common to all RETs, declined drastically from 2010 to 2024, falling 93% from US\$2571/kWh to US\$192/kWh (International Renewable Energy Agency (IRENA), 2025). Although TIC for Solar PV has fallen significantly in recent years, the total projected TIC for solar or other RETs to electrify the large number of non-electrified households in MHS Province still greatly exceeds available budgets and represents a key barrier.

Ec2(S): Solar PV TIC. Solar PV systems have experienced a dramatic 87% decrease in TIC since 2010 and have the lowest cost per kW of the four technologies (International Renewable Energy Agency (IRENA), 2025). However, the costs to provide a stand-alone community solar PV project in the remote areas of MHS Province are still significant and the small projects provide a low return on investment when considering the number of households served. For example, PEA's current Community Solar Pilot Project will provide community solar PV systems for five villages and 301 households in MHS's Khun Yuam, Mae Sariang, and Mae La Noi districts at a total cost of 83,500,000 baht (US\$2.53 million), which equates to 277,409 baht (US\$8406) per household (Provincial Electricity Authority, Mae Hong Son Province Office, Thailand, 2024).
Ec2(W) Wind power TIC. Land-based wind power systems have experienced a 55% decrease in TIC since 2010. Wind power is a viable technology for rural electrification (International Renewable Energy Agency (IRENA), 2025) and has a relatively low cost per kW. However, the small, low wind speed turbines appropriate for rural MHS Province are less efficient and less cost effective. The remoteness of the villages increases installation costs in transporting the turbine and other associated components, building the foundation and tower, and employing a crane (Roynarin, 2024; Thunya Power, 2012).
Ec2(B): Bioenergy TIC. Bioenergy electrical generation TIC is about 4.7 times as expensive per kW than solar PV (International Renewable Energy Agency (IRENA), 2025) and is not cost effective for MHS Province where biomass availability is also a limiting factor (Chansopa, 2023). The IRENA study considered all biofuels such as wood chips, pellets, and bagasse, as well as renewable municipal waste (International Renewable Energy Agency (IRENA), 2025). Also, commercial companies operate bioenergy plants in Thailand and low return on investment presents a significant barrier, particularly in rural areas such as MHS (Fupol, 2024).
Ec2(H): Hydro TIC. Small/micro-hydro project costs include procurement of expensive equipment with high design and construction costs, and TIC has increased by 52% since 2010 (International Renewable Energy Agency (IRENA), 2025). One issue with hydro projects is that each installation requires a custom design based on the unique characteristics of the local water flow and installation area. In contrast, solar PV projects can be constructed in cookie-cutter fashion with each project being almost identical (Greacen, 2023). Due to the falling prices of solar PV components, hydro projects are no longer economically competitive.
Ec3: Levelized cost of electricity (LCOE) of RETs. Table 4 provides IRENA's LCOE for the four RETs relevant to MHS Province. LCOE is the average cost of generating electricity over the lifetime of the project and includes additional investments as well as operations and maintenance costs from the time the project begins operation until decommissioning. The four RETs are individually addressed below.

Ec3(S): Solar PV LCOE. Solar PV LCOE has the second lowest LCOE just above land-based wind power systems and has experienced a very significant 90% decrease in cost since 2010 (International Renewable Energy Agency (IRENA), 2025), but the total costs for electrification still significantly exceed available budgets.
Ec3(W) Wind power LCOE. After installation, wind turbine systems have the lowest LCOE of the four technologies (International Renewable Energy Agency (IRENA), 2025), but the costs are still significant. The costs are mainly for turbine maintenance and periodic bearing replacement (Roynarin, 2024). Although IRENA's costs reflect large-scale wind turbine projects, they are assumed to roughly apply to small wind turbines applicable for small village projects.
Ec3(B): Bioenergy LCOE. Bioenergy systems have the highest LCOE at double that of solar PV [5050]. Also, specialized bioenergy system maintenance experts are not available in remote rural provinces like MHS, so maintenance costs can be expected to be even higher.
Ec3(H): Hydro LCOE. Hydro system LCOE is similar to LCOE for solar PV (International Renewable Energy Agency (IRENA), 2025). However, historically, the small village systems installed by DEDE experienced many failures due to the significant seasonal variations in stream water flow. Excessive water flow in the rainy season between June and October results in damage to turbines due to the amount of

debris in the water, incurring costs to repair the damage (Khamthiya, 2023).

Ec4: Low income of residents. Individual home solar PV installations paid for by homeowners are an option for residents in non-electrified villages. Although solar PV system costs have fallen dramatically, these residents typically have low incomes (National Statistics Office (Thailand), 2025b) and cannot afford individual personal systems (Dihae, 2023; Khamthiya, 2023; Ninudomsak, 2023; Panchoo, 2023; Saisord, 2023; Wongprai, 2024).

Ec5: Low/no government subsidies. There are currently no government-provided subsidies to help private citizens purchase solar PV systems for their homes, preventing less affluent families from being able to afford the systems (Ninudomsak, 2023; Wongprai, 2024).

Ec6: Limited financing options. It is extremely difficult for poor rural farmers in MHS Province to obtain bank loans to purchase solar PV systems due to their low and inconsistent income and lack of collateral. Government loans, microfinancing options, and crowd-funding are either not yet available or too difficult to obtain. For many poor residents, the only options are illegal black market loans with extremely high interest rates or to save money and pay in cash (Dihae, 2023; Khamthiya, 2023; Ninudomsak, 2023; Saisord, 2023; Wongprai, 2024). Although Thai law caps loan interest at 15% annually (Lortrakul, 2024), illegal black market loans can be as high as 15–20% monthly (Dihae, 2023; Khamthiya, 2023; Ninudomsak, 2023; Saisord, 2023; Siam Commercial Bank, 2020; Wongprai, 2024).

Ec7. Lack of funding for system life cycle costs. Past projects have often lacked additional funding for operations and maintenance after installation is complete. This has been especially true of smaller systems provided by non-governmental organizations (NGOs). Eventually repairs are needed, such as replacing failed inverters or lead-acid batteries that have reached the end of their life. Project recipients lack funds to make repairs, so the broken systems sit idle (Dihae, 2023; Khamthiya, 2023).

Social barriers to rural electrification

Eight current social barriers to MHS Province's rural electrification were identified:

S1: Public perception that electricity from RETs is less reliable than power from the main grid. A public perception is that energy from renewable sources is less reliable than energy from the main grid. This results in some resistance to new RET-based electrification projects, as explained by a PEA subject matter expert:

"Some people don't trust renewable energy; they want to get electricity from the power lines. Education is the key to overcome this thinking."

PEA Project Manager for the MHS Community Solar Pilot Project (Boonsiri, 2023)

However, newer community solar systems have recently proven to be a more reliable source of electricity than grid-provided power, which in MHS Province experiences frequent outages (Chansopa, 2023; Ninudomsak, 2023; Panchoo, 2023; Wailom, 2023). As more community solar projects are successfully installed and operated, this perception is lessening. The three villages in Huay Pu Ling Subdistrict which received community solar systems in 2022 have reported zero outages or days when the energy storage was depleted since the systems started providing electricity (Denpraipana, 2024; Jehae, 2024; Praiprasertying, 2024; Saengcherdfa, 2024). This stands in sharp contrast to the many outages experienced by MHS province's main grid, which also lessens the impact of this barrier.

S2: Public perception that opting for community PV will preclude future main grid extension to a village. Another public perception is that if a community opts for a stand-alone community PC solar system to be built, they will never get the grid extended to their village (Tungpokanon, 2023). Because of this concern, some LAOs have opted to request funding for other civic projects while they wait for an extension of power lines from the main grid, which may never be approved due to permitting issues with forest encroachment and the low return on investment of extensions to small communities.

S3: Difficulties with local management of electricity cooperatives. Stand-alone community renewable energy systems remain under the management of the installing agency (usually an MOE department, PEA, or the provincial government) for a period of only two years before full control and responsibility are passed to the village. These responsibilities include system maintenance and repair as well as collecting usage fees from households and managing the co-operative's finances. This policy can result in community resistance, as explained by the Province Energy Director:

"*Villages prefer power lines not renewable energy because it is easier for them, PEA maintains the power, villagers just pay the bill. With renewable energy the village must set up a committee and there are complex maintenance issues.*"

MHS Province Energy Director (Wailom, 2023)

S4: Shortage of skilled technical manpower. MHS Province has a limited number of residents with the necessary technical training, experience, and qualifications to effectively install and maintain complex electrical systems. This barrier is partially due to the province's shortfalls in the education and training of its residents, which increases the difficulty for local villagers to maintain community energy systems that fall under their responsibility after the initial two-year post-installation period. Many of the DEDE community micro hydro systems built over the past 30 years are now inoperative as the local villages lack the expertise, as well as the finances, to repair them (Phunphaeng, 2023). UNDP ranks MHS below the national average for all educational-related SDG indicators (United Nations Development Programme (UNDP), 2024).

S5. Lack of higher education opportunities. MHS Province has only one institution offering a four-year degree, Chiang Mai Rajabhat University, Mae Hong Son College, but with no energy-related areas of study. Two of the three two-year community colleges in MHS province, Nawamintrachinee Mae Hong Son Industrial and Community Education College in MHS municipality and Mae Sariang Industrial and Community Education College in Mae Sariang District, offer vocational education as well as specialized instruction in solar PV systems (Mae Sariang Industrial and Community Education College, 2025; Nawamintrachinee Mae Hong Son Industrial and Community Education College, 2025). The province's limited education opportunities present a barrier to having a local workforce with the ability to operate and maintain energy projects in the province.

S6: Lack of recognition and rights for stateless people. According to the MHS Provincial Development Plan for 2023–2027, there are 29,548 stateless individuals in the province who have applied for Thai citizenship. Out of this number, only 1038 individuals have already been granted citizenship, leaving 28,510 individuals still awaiting approval as of 2023 (Secretary of the Provincial Administrative Organization, Mae Hong Son Province, Thailand, 2024). Many of these stateless people currently live in non-electrified villages and face additional electrification barriers such as the inability to own land.

S7(H): Competition for water resources between hydro projects and agriculture. During the dry season between November and June, hydroelectricity systems compete for the limited water supply with farmers who need the water to irrigate their crops. Ultimately, the need to grow food and feed the community is a higher priority than producing electricity (Dihae, 2023; Khamthiya, 2023). This conflict has often been cited as an issue with small village hydro power projects, such as in Khun Pae village in Chiang Mai (Wongprai, 2024). This barrier, along with other issues with hydro projects, has resulted in the abandonment of many DEDE hydro systems constructed in the past 30 years.

S8: Desire to maintain the natural beauty of the province. MHS Province's leaders and residents have a strong desire to maintain the natural beauty of their province. The building of transmission lines even within a village boundary can be seen as disturbing the natural beauty, which is important to many villagers' way of life. Because of this, there can be some resistance to all development projects, including those that bring electricity to a village (Onwong, 2023; Panchoo, 2023).

S9: Land usage conflicts. Although MHS Province is sparsely populated with available land for most electrification projects, poor coordination with village residents can lead to land use conflicts and resistance to the project. Planned projects can in some cases interfere with agricultural activities causing conflicts. One potential solution to this barrier is an agrivoltaic system, a dual land-use technique where crop growing and energy production can be conducted in the same agricultural field in a symbiotic manner (Chopdar et al., 2025; Giri & Mohanty, 2024). Floating solar photovoltaics (floatovoltaics) is another possible solution. Both technologies will be examined in future research.

Technological barriers to rural electrification

Six current technological barriers to MHS Province's rural electrification were identified:

T1: Low quality of solar PV systems. Past renewable energy systems suffered from a lack of quality, resulting in system failures. This was particularly true of the SHS project, which installed SHSs in about 14,800 homes in MHS Province in the mid-2000s during the Thaksin Shinawatra administration. Research by the French Development Agency (Agence Française de Développement) estimated that by 2007, as many as 80% of the 300,000 SHSs installed in the entire country had become inoperative (United Nations Development Programme, 2009). Although today's consumer solar PV systems feature better reliability, the low-cost systems chosen by many consumers with limited budgets feature components with high failure rates due to poor quality control and low-quality components (Wongprai, 2024; Yinsinsomboom, 2024).

T2: Lack of cellular network coverage. Many remote areas of MHS Province are out of the range of cellular networks, making coordination difficult for installation and maintenance crews. Mobile phones are also often used as Global Positioning System (GPS) receivers during surveying (Wailom, 2023). For example, almost all Huay Pu Ling Subdistrict (one of the least electrified subdistricts in the province), lacks cellular network coverage.

T3(B): Bioenergy systems require maintenance and can be more complex and costly to repair. Although a large variety of bioenergy technologies ranging from using various crop residues to animal and municipal waste have been used in rural areas of the developing world for decades (Díaz González & Sandoval, 2020; Luo et al., 2018), these systems have limitations that create barriers for their use. All bioenergy sources involve a conversion from biological material to energy. Technical issues with this technology include high maintenance costs, inadequate and improper maintenance practices, limited availability of spare parts, and lack of skilled technicians leading to a failure to follow maintenance and operations procedures (Allesina & Pedrazzi, 2021; Barry et al., 2021; Chambon et al., 2020; Owen & Ripken, 2017; Palit et al., 2013). As a result, some bioenergy installations in rural areas are no longer operational (Iqbal et al., 2020; Whitty, 2022). These technical barriers can combine to make bioenergy technologies less suitable in rural areas where there is a lack of technical expertise.

T4(B): Lack of biomass for bioenergy production. Due to the small population and limited agriculture, MHS Province has low energy potential for solid biomass primarily from unused components from rice, maize, and soybeans (Department of Alternative Energy Development and Efficiency, Ministry of Energy, Thailand, 2023). Also, bioenergy faces an obstacle from the trade-off between growing crops for food security or for cash crops versus biomass production (Díaz González & Sandoval, 2020; Luo et al., 2018; Owen & Ripken, 2017).

T5(W): Lack of information on wind conditions. While there is high-level satellite derived data on wind conditions available for MHS Province (Department of Alternative Energy Development and Efficiency, Ministry of Energy, 2025), there have been no in-depth wind surveys, especially in mountainous rural areas. It's likely that there are greater wind speeds at higher elevations, particularly in mountain passes or on ridges. However, until more detailed wind surveys are conducted, the potential for wind power production will not be known (Roynarin, 2024).

T6: Initial capacity estimates for villages are low and do not account for growth. The community PV systems installed in MHS Province since 2022 are smaller systems with modest output. Although the systems have not yet encountered issues with capacity, village leaders have acknowledged that villagers are saving to purchase more electrical appliances (e.g., washing machines) in the future (Denpraipana, 2024). As there are no plans to increase the output of the solar arrays, this barrier looms as a future issue for many of these communities reliant on stand-alone community solar PV.

Environmental barriers to rural electrification

Nine current environmental barriers to MHS Province's rural electrification were identified:

En1: Mountain roads cause accessibility issues. Roads in the mountainous areas of MHS Province, where the bulk of the non-electrified villages are located, are narrow, often unpaved, and feature steep inclines (Secretary of the Provincial Administrative Organization, Mae Hong Son Province, Thailand, 2024). These dirt roads become impassable for weeks at a time, even for off-road vehicles, during the rainy season due to mud. Transportation of equipment for either renewable systems or traditional power lines is slow, difficult, costly, and sometimes impossible. The mountain roads also pose a barrier to the potential transport of biomass materials, whether agricultural products or animal waste, from mountain farms to a bioenergy plant.

En2: Mountainous terrain increases the costs of constructing electrical distribution lines. The mountainous topography of most of MHS Province also imposes economic barriers to electrification via traditional transmission lines, such as PEA's lines from Chiang Mai Province. Even when following the paved main highways, the many curves and switchbacks make construction more difficult, increasing the cost of construction (Chansopa, 2023).

En3(W): Relatively low wind speeds. Although detailed studies have not been conducted, in general, the MHS Province experiences relatively low wind speeds, typically Class 5 winds. These winds are on the order of about 5 m/s, which can be inefficient for traditional wind turbines (Department of Alternative Energy Development and Efficiency, Ministry of Energy, 2025; Roynarin, 2024).

En4(W): Optimum wind turbine locations are far from village mini grids. Related to Barrier T5(W), any future wind turbine installation in MHS Province would likely be at higher elevations where there are typically higher wind speeds, resulting in increased power output. But in the case of community systems, this places the turbines at a

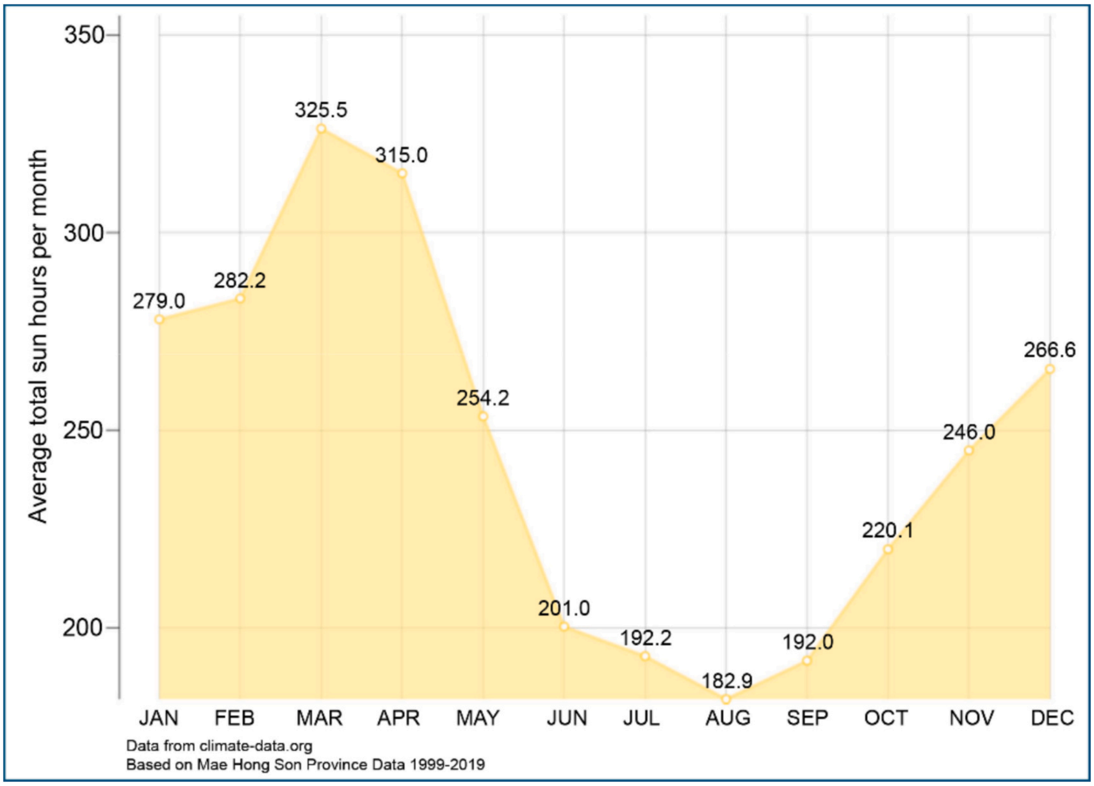

Fig. 3. Average total sun hours per month for MHS province.
Adapted from https://en.climate-data.org/asia/thailand/mae-hong-son-province/mae-hong-son-3729/ (n.d.).

distance from the community PV array, which is typically located in lower-lying areas in or near the village. The distribution lines from the turbines to the village mini grids would require the construction of power poles and the necessary permits to allow for their installation (Roynarin, 2024).

En5(S): Atmospheric conditions filter solar radiation. The “three mists” of MHS Province – clouds in the rainy season, smoke in the dry season, and fog in the cold season– all reduce the solar energy reaching PV cells and thus reduce PV output. Cloud cover and fog are the most significant factors with heavy cloud cover accounting for as much as a 67% loss in PV power output (Amusan & Otokunefor, 2023).

Weather plays a significant role in MHS Province's solar electrification situation. Cloudy sky conditions during the rainy season from June through October result in decreased electrical production from solar PV systems due to limited solar radiation reaching the PV panels. Fig. 3 depicts the average daily number of hours of sun each month in MHS Province based on European Centre for Medium-Range Weather Forecasts (ECMWF) climatology data from 1999 to 2019 (https://en.climate-data.org/asia/thailand/mae-hong-son-province/mae-hong-son-3729/, n.d.). Cloud cover presents a significant barrier as solar PV is a leading electricity production source for remote villages.

En6(S): Regionally produced aerosols from agricultural burning reduce solar PV output. The 7SEAS study by Reid et al. (2013) described how particulate matter from agricultural burning across the northern Indochina region can reach high levels during the burning season from December to May. The monsoon-driven winds during the winter provide the transport mechanism for aerosols from Myanmar over northern Thailand and MHS Province (Reid et al., 2013). Although Thailand is taking steps to reduce the seasonal burning of crops, fires in neighboring Myanmar significantly reduce the effectiveness of local measures. EGAT reported that the power output of their Pha Bong solar PV project decreases by about 10% during the burning season on days when PM2.5 concentrations are high (Chaipruek, 2023).

En7(H): Insufficient water flow for hydro in the dry season. Many of the hydro systems constructed in the past 30 years by DEDE have been abandoned, often due to insufficient water flow to operate during the dry season between November and June (Dihae, 2023; Khamthiya, 2023).

En8(H): Excessive water flows in the rainy season. During the rainy season, water flow in the streams is sufficient to produce power from hydroelectric sources, but large rainstorms can bring excessive rainfall, causing large amounts of debris to flow into the turbine. The debris is mostly from natural sources, such as branches and other vegetation, washed away by the stream during storms. This debris can cause turbine damage, and the communities lack the funding and technical expertise to carry out repairs, resulting in broken and inoperative hydro systems (Khamthiya, 2023).

En9: Widespread Watershed Class 1 and Class 2 areas. Much of MHS Province, and especially the non-electrified villages, are located within Watershed Class 1A, 1B, and 2 areas. As part of the permitting process, any projects that impact land designated as Watershed Class 1 require conducting an Environmental Impact Assessment (EIA), a costly and time-consuming process (Boonsiri, 2023). Even with the EIA completed, the RFD and the even stricter DNP have not historically approved permits for construction in Watershed Class 1 areas (Chansopa, 2023).

Legal barriers to rural electrification

Four current legal barriers to MHS Province's rural electrification were identified:

L1: Obtaining permits for projects on protected land is extremely difficult. The single most difficult barrier to 100% electrification in

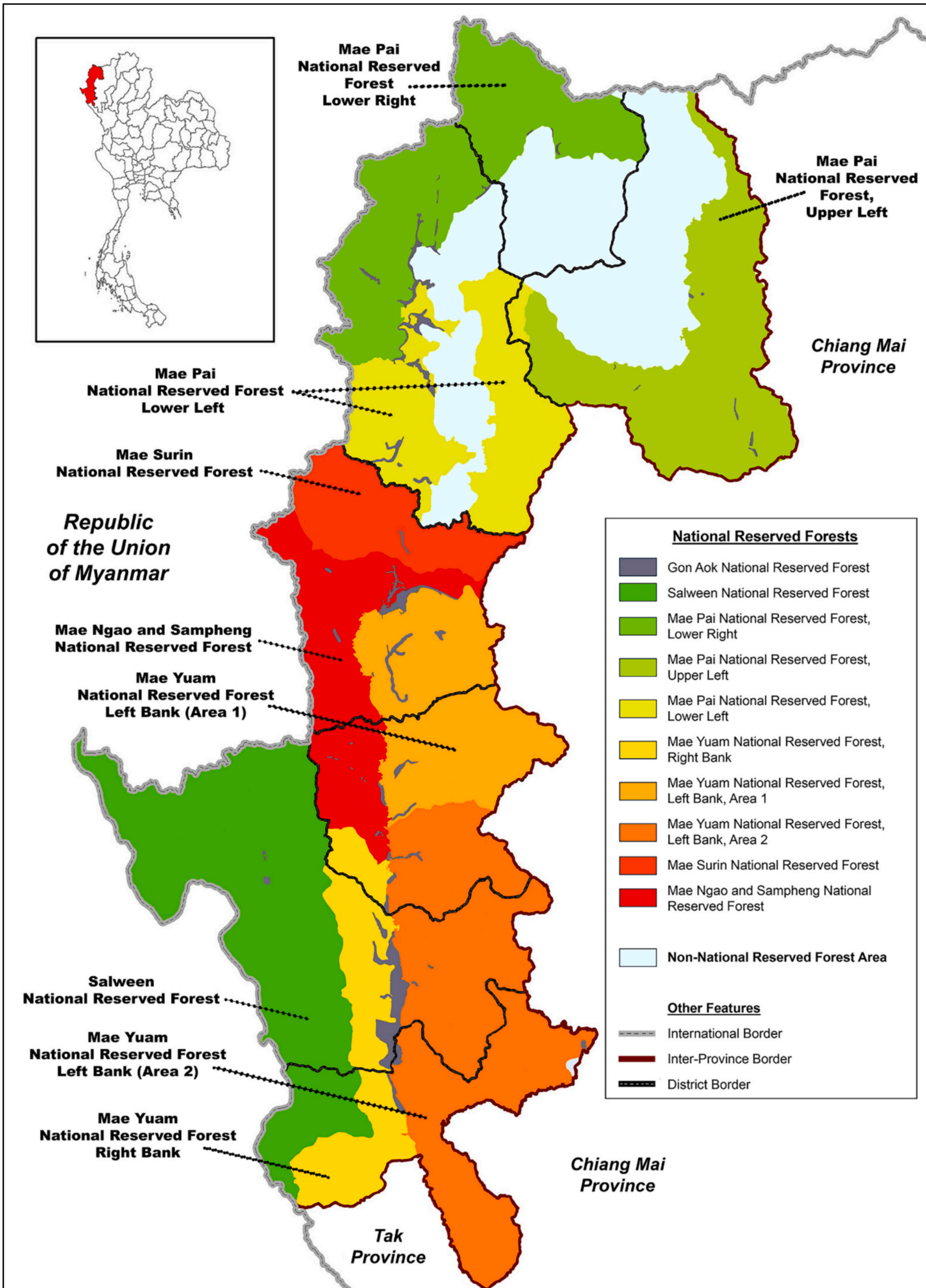

Fig. 4. National reserved forests in MHS Province.
Adapted from Mae Hong Son Province (2024a).

MHS Province is obtaining the required permits to conduct any development on protected land, including any electrical projects with transmission line construction. Currently, all 92 non-electrified main villages are on protected land of some sort (Mae Hong Son Province Energy Office, Ministry of Energy, 2024). The permits are not only required for long-distance transmission line construction through the forests and national parks but are also required for any construction within the villages (Wailom, 2023). Obtaining these permits was universally cited as the most difficult barrier by the subject matter experts who were interviewed (Boonsiri, 2023; Chansopa, 2023; Panchoo, 2023; Tungpokanon, 2023; Wailom, 2023).

L2: EIA requirements. In addition to the EIA requirements for lands in Watershed Class 1 and Class 2 areas described in Barrier En9, permit requests on land in a Conservation Forest Zone within a Reserved Forest also require conducting an EIA (Chainarinphat, 2025a). EIAs are also required if the affected area contains rare plants or rare animal species. For example, MHS Province's old teak forests and a rare moss that grows near Thai Rak Village both have special protections (Charoensuepsakul, 2025a). EIAs typically take two years. The first year is for data collection and report preparation as the area must be studied in all three seasons. The second year is typically spent in government review before approval (Chansopa, 2023).

L3. Difficult wind power system permitting. For wind power systems, permits are required to install wind turbines and transmission lines to carry the electricity from the turbine generator, adding to the overall cost and acting as a barrier for wind turbine construction (Roynarin, 2024). Permits are also required to install wind monitoring systems on land under the jurisdiction of the RFD and DNP (Chansopa, 2023).

L4. Lack of land ownership by Hill Tribe residents. Many MHS Province ethnic Hill Tribe residents reside in homes and villages on land they do not legally own, which are within the borders of legally protected forest areas. This is despite the fact that many of these people have lived on these lands for generations, well before the lands were designated as protected. This serves as a barrier to permitting and construction (Chansopa, 2023; Wailom, 2023).

Discussion

The 39 barriers identified by this research all contribute to some degree to the large number of non-electrified villages in MHS Province. However, three key barriers emerged as the most salient from a qualitative analysis based on the combination of their frequency across subject matter expert interviews, the consistency of emphasis across different stakeholder groups, and their alignment with well-established barriers reported in the rural electrification literature. The three key barriers are listed below.

- Lack of funding (Ec1)
- Village locations in remote mountainous areas (En2)
- Difficulty obtaining permits for electrification projects (L1).

Lack of funding (Ec1) was consistently mentioned across nearly all stakeholder categories and is widely recognized as a primary constraint in rural electrification efforts. If funding were available, full electrification could be accomplished quickly. But budget constraints and the high cost of the projects (as described in barriers Ec2 and Ec3) limit how many villages can be electrified each year.

Village locations in remote mountainous areas (En2) emerged as a significant barrier due to repeated references to transportation difficulties, increased infrastructure costs, and logistical challenges, all of which are well documented barriers in prior studies. Both grid extensions and stand-alone projects featuring RETs yield a small return on investment as the installations are costly for the small number of households serviced.

Multiple subject matter experts emphasized the difficulty in obtaining permits for electrification projects (L1) as the most constraining barrier to electrification in MHS Province. The mountain villages and their residents face a unique combination of factors which make permitting extremely difficult.

> "*Along with the budget the biggest barrier is to get permission from the Forest Department.*"
>
> MHS Province Energy Director (Wailom, 2023)

> "*Getting permits is most difficult part of the project.*"
>
> PEA Project Manager for the MHS Community Solar Pilot Project (Boonsiri, 2023)

> "*The biggest barrier is the village is on Forest land and getting permission from the Forest Department is a confusing, long process that takes many years.*"
>
> Mueang MHS Deputy District Chief (Panchoo, 2023)

> "*We have some budget, but it is very difficult to get a permit.*"
>
> Director, MHS PEA Office (Chansopa, 2023)

> "*Funding for village electrification is less of an issue than getting a permit.*"
>
> MOE Plans and Policy Analyst, Expert Level (Tungpokanon, 2023)

The factors underlying barrier L1 in MHS Province are unique to the region and are discussed further below.

Archeological evidence supports the estimation that Hill Tribe ethnic groups first settled in these locations in the 14th or 15th century in the present-day towns of Pai, Mae Sariang, Mae Hong Son, and Khun Yuam. Since that time, there has been a near continuous migration into the region by various ethnic peoples from the Shan, Hmong, Muser, Lisu, Lawa, and other tribes (Pradit, 2019). These people tended to settle in MHS's higher elevation villages and so are known as the Hill Tribes (Mae Hong Son Province, 2025b) who today comprise just over 50% of the province's population (Mae Hong Son Provincial Social Development and Human Security Office, Thailand, 2020). In most cases, the Hill Tribe people living in the mountainous villages lack titles for the land their families have lived on for many generations.

Beginning in 1941 with the Forest Act, B.E. 2484, the Thai government passed a series of laws in the latter part of the 20th century designating much of the country's sparsely populated land as protected lands. Thai law designates any land which has not been legally acquired as "Permanent Forest" (*Forest Act, B.E.* 2484, 1941), with 85.9% of MHS Province's total land area of 12,780 km2 falling under this designation. Within Permanent Forests are areas designated by the National Reserved Forest Act of 1964 as "National Reserved Forests", which are forest lands reserved to conserve the forest condition, forest products, or other natural resources (*National Reserved Forest Act, B.E.* 2507, 1964). Fig. 4 illustrates the extent of the National Reserved Forest areas in MHS Province (Mae Hong Son Province, 2024a). Permanent Forests and National Reserved Forests fall under the jurisdiction of the RFD.

Thailand's National Forest Policy includes the stated objective to "stop and prevent the destruction of forest resources in all forms of state forest land". The document's list of causes of forest area reduction includes "infrastructure development" and specifically cites electricity distribution, in other words the construction of electrical transmission lines (Ministry of Natural Resources and Environment, Thailand, 2020). These policies make it extremely difficult for Thailand's energy agencies such as PEA or EGAT to obtain approval for energy projects on legal forest land, whether for extending the main grid or constructing RET projects in villages.

Overlapping the DNP's protected forests are additional protected areas under the jurisdiction of the DNP, also under the Ministry of Natural Resources and Environment. The DNP's protected areas include National Parks, Forest Parks, Botanical Gardens, Wildlife Sanctuaries,

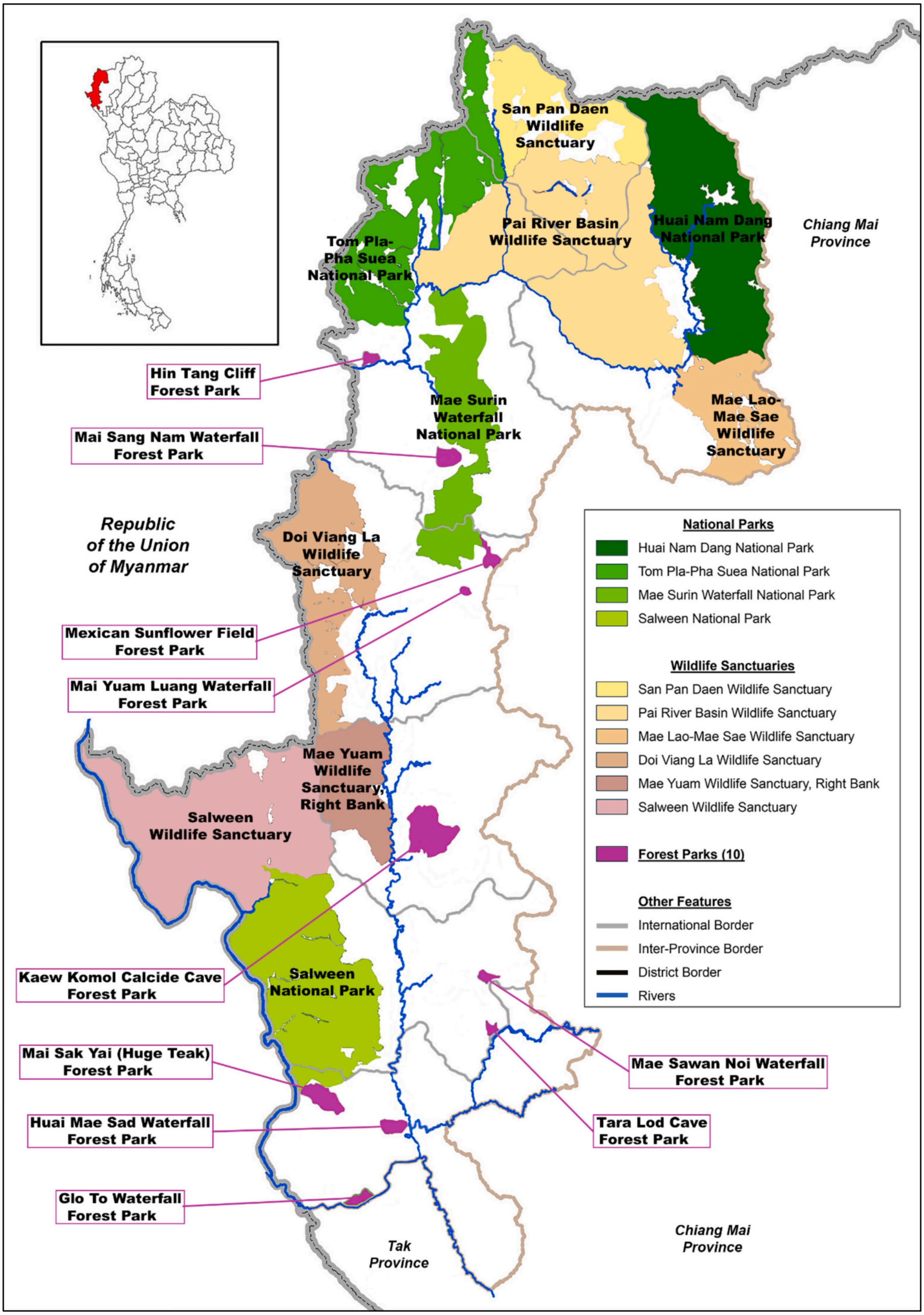

Fig. 5. National parks, forest parks, and wildlife sanctuaries in MHS Province.
Adapted from Mae Hong Son Province (2024a).

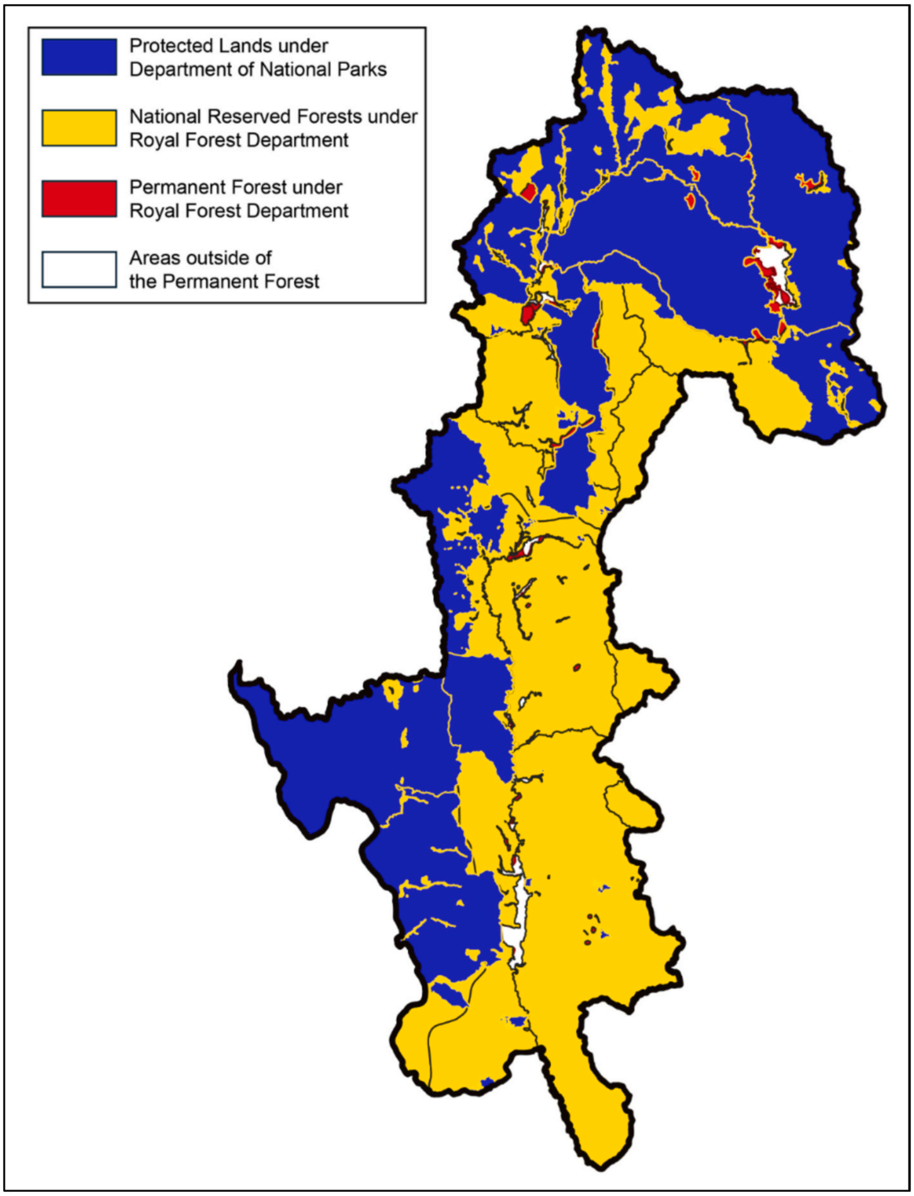

Fig. 6. Jurisdiction of protected lands within MHS Province.
Adapted from Royal Forest Department, Mae Hong Son Office, Thailand (2025).

and Non-Hunting Areas as defined by the National Parks Act of 2019 (*National Parks Act, B.E.* 2562, 2019) and the Wild Animal Conservation and Protection Act of 2019 (*Wild Animal Conservation and Protection Act, B.E.* 2562, 2019). Fig. 5 illustrates these areas in MHS Province (there are no Botanical Gardens or Non-Hunting Areas) which occupy 6696 km2 (52%) of the province's land area (Mae Hong Son Province, 2024a).

Figs. 4 and 5 illustrate the significant overlap between the protected lands managed by the RFD and DNP. Prospective projects located in both an RFD and NPD-managed area will fall under the jurisdiction of only one of the two departments, usually defaulting to the DNP's jurisdiction (Chainarinphat, 2025b). This results in the combined map shown in Fig. 6 (Royal Forest Department, Mae Hong Son Office, Thailand, 2025).

Table 5
Land designations for non-electrified villages in MHS Province. Compiled by the authors based on Mae Hong Son Province Energy Office, Ministry of Energy (2024).

Land type	Main villages	Subvillages	Total villages
Land title deed	0	1	1
Permanent forest	0	2	2
National reserve forest	73	158	231
National park	4	22	26
Wildlife sanctuary	15	29	44
Total	92	212	304

In general, it is more difficult to obtain development permits from the DNP than the RFD, as the RFD provides some leeway to allow for economic improvement for residents (Fuengfoo, 2023). The DNP has the duty and responsibility to preserve their protected lands without economic considerations (Charoensuepsakul, 2025b). If a public project is less than 50 rai (19.8 acres/80,000 m2) in size, permits are approved at the local level by the offices in MHS. If a project is over 50 rai in size, final approval must come from the Ministry of Natural Resources in Bangkok, which adds significant delay. Adding to the difficulty, the permitting process for the two departments are different and both are 100% paper based, requiring the routing of hard copy forms through the entire approval process (Charoensuepsakul, 2025b; Fuengfoo, 2023). Individual residential solar PV installations for homes in legal forests do

Table 6
Watershed classification and characteristics.

Watershed class	Slope (%)	Characteristics
Class 1	>50	High elevation areas in the upper part of the mountainous regions with steep slopes, valleys, cliffs, and many watercourses that are the headwaters for streams.
Class 2	35–50	High elevation areas in the upper part of the mountainous regions with rounded ridges and moderately long slopes, fairly wide canals, and cleared rainforests or degraded forests.
Class 3	25–35	Upland or foothill areas with moderate to steep slopes, less prone to erosion than Watershed Class 1 and 2 areas.
Class 4	6–25	Foothill areas along the banks of rivers or streams. These areas have generally already been encroached upon and deforested for agriculture or other activities.
Class 5	<6	Areas that are almost flat or have gentle slopes. Forests in these areas have been cleared for agriculture, especially rice paddies and other crops.

Adapted from Royal Forest Department, Ministry of Natural Resources and Environment, Thailand (2009).

not require permits, but projects for businesses must first obtain approval (Chainarinphat, 2025b).

Under various Thai laws, it is illegal for villagers to live and farm in any of these protected areas, apart from those whose occupation or use predates the establishment of the protected area and who have formally applied for an exemption. However, any development including transmission line extensions or community RET projects must receive permits from the RFD or DNP. The difficulty in obtaining a permit from RFD or DNP has been consistently cited as the most difficult barrier to electrifying the remaining nonelectrified villages in the province.

Table 5 provides the legal land designation breakdown for each of the 304 non-electrified main villages and subvillages in MHS Province with 100% of the non-electrified main villages and all but one of the 212 non-electrified subvillages located on legally protected land. Only the residents of Huai Rai subvillage in the Mueang District hold titles to the land they live on (Mae Hong Son Province Energy Office, Ministry of Energy, 2024).

Overlaying the complex system of RFD and DNP protected areas, Thailand also designates Watershed Areas under the management of the Office of Natural Resources and Environmental Policy and Planning to control and minimize impacts on water quantity, quality, and flow duration (Office of Natural Resources and Environmental Policy and Planning, Ministry of Natural Resources and Environment, 2025). A watershed is defined as a geographic area related to water management, and the five classes shown in Table 6 relate to the area's environmental fragility and susceptibility to soil erosion (Royal Forest Department, Ministry of Natural Resources and Environment, Thailand, 2009).

Watershed Classes 1 and 2 have the highest degree of protection, with Watershed Class 1 divided into two subclasses, 1A and 1B. Class 1A watersheds are those areas where the forest has been classified as undeveloped, while in Class 1B watersheds the forest has previously been cleared for development (Royal Forest Department, Ministry of Natural Resources and Environment, Thailand, 2009). Due to its predominately mountainous terrain, a significant number of MHS Province's villages are currently located within Watershed Class 1A, 1B, and 2 areas, as shown in Table 7 (Mae Hong Son Province Energy Office, Ministry of Energy, 2024) and Fig. 7 (Mae Hong Son Province, 2024a). 11,544 of MHS Province's total 12,780 km^2 (90.3%) are in Watershed Class 1A, 1B, or 2 lands (Mae Hong Son Province, 2024b).

Development in Watershed Areas is a concern for the RFD, not the DNP (Charoensuepsakul, 2025b), unless the land is also designated as a National Park, Forest Park, or Wildlife Sanctuary. Thai law prohibits any changes to the characteristics of Class 1 or Class 2 forest areas to preserve the watershed. Exceptions must be granted at the Cabinet level. A recent Cabinet resolution on 25 February 2025 approved a waiver for PEA to extend electricity distribution lines to 29 non-electrified main and subvillages in the Pang Mapha and Pai Districts of northern MHS Province near the Myanmar border. Approval of this 337-million-baht (US $10.2 million) project is a significant development as it includes the construction of over 70 km of electricity distribution lines passing through Class 1A Watershed Areas, as well as designated National Reserved Forest, National Park, and Wildlife Sanctuary areas. The project falls under the Border Security Development Village Project, which was initiated by a Royal Decree in 2012. The 13-year timeline to achieve Cabinet approval illustrates the difficulty of the permitting process, even with a royally sponsored project (Ministry of the Interior, Thailand, 2025). The fact that 61.8% of MHS's non-electrified main and subvillages are contained in Watershed Class 1A or 1B areas presents a significant barrier to electrifying these villages. Overall, the situation with most MHS provinces' non-electrified villages being in restricted areas is repeatedly cited as the most significant barrier to electrification by the many experts who were interviewed as part of this research.

Table 8 presents the practical recommendations resulting from this research for each of the six PESTEL dimensions.

Conclusion

Summary of major findings and contextual insights

This study identified 39 interrelated barriers that continue to impede progress toward universal rural electrification in MHS Province, Thailand. Using a structured PESTEL-based qualitative framework that integrates document analysis with expert stakeholder interviews, the findings demonstrate that electrification challenges in MHS extend well beyond technical and financial constraints. Instead, they are embedded within complex social, environmental, legal, and institutional contexts.

While some barriers—such as limited funding, remoteness, and challenging topography—are common across rural regions globally, MHS Province exhibits context-specific characteristics that intensify these challenges. These include long-established Hill Tribe communities lacking formal land tenure, extensive forest conservation designations, and highly restrictive regulatory regimes. Additionally, uncontrollable environmental factors such as weather variability and climatological conditions constrain the performance of solar and wind energy systems. Collectively, these conditions create prolonged approval timelines and

Table 7
MHS Province villages in watershed class areas. Compiled by the author based on Mae Hong Son Province Energy Office, Ministry of Energy (2024).

Watershed class	All villages				Non-electrified villages			
	Main	Sub	All	Percent	Main	Sub	All	Percent
1A	74	130	204	25.9%	36	93	129	42.4%
1B	44	54	98	12.5%	15	38	53	17.4%
2	63	71	134	17.0%	17	42	59	19.4%
3	107	79	186	23.6%	22	36	58	19.1%
4	88	20	108	13.7%	2	3	5	1.6%
5	45	12	57	7.2%	0	0	0	0.0%
Total	421	366	787		92	212	304	

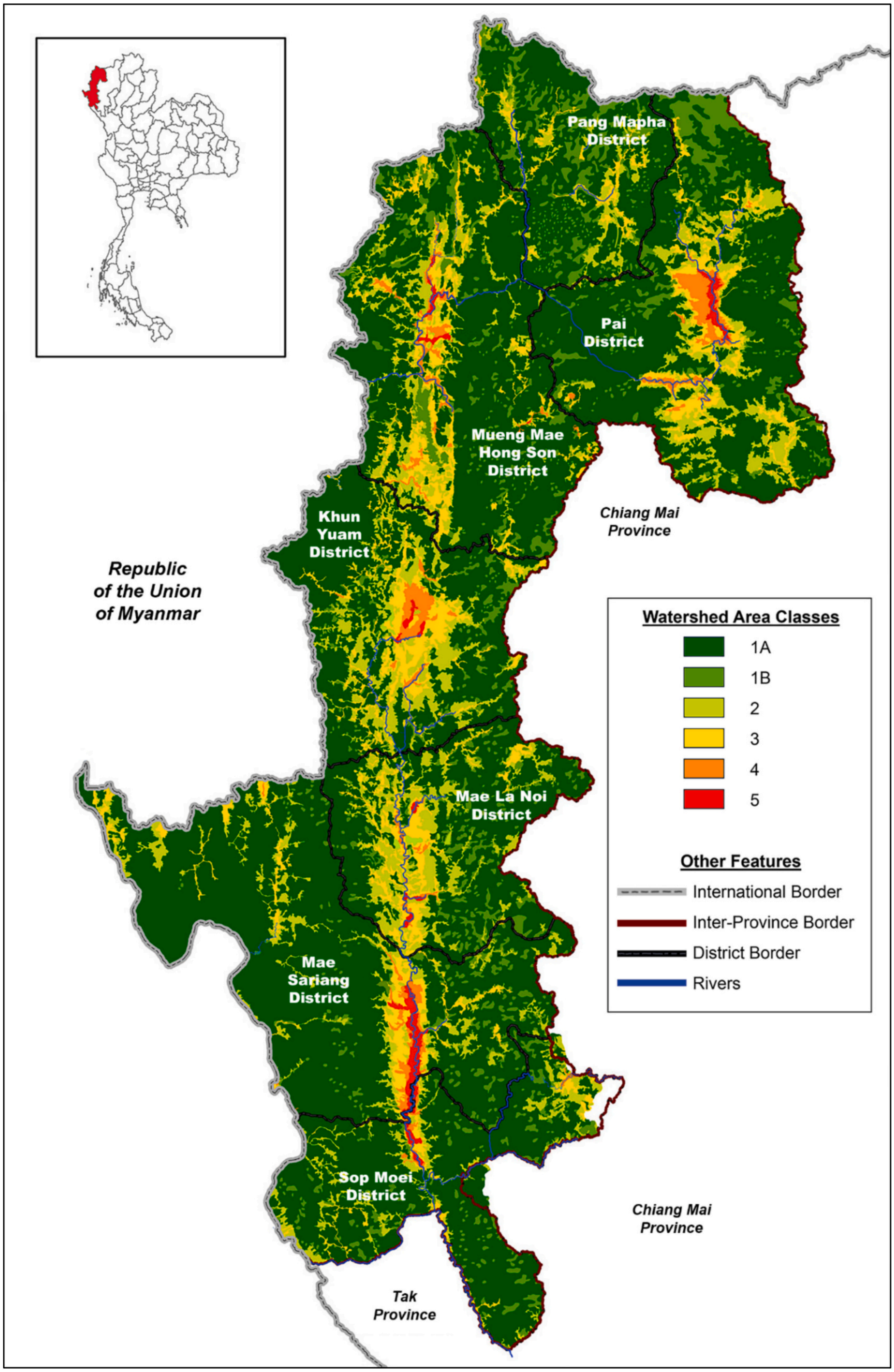

Fig. 7. MHS Province watershed map.
Adapted from Mae Hong Son Province (2024a).

Table 8
Practical recommendations by PESTEL dimension.

Dimension	Key barriers identified	Recommended actions/policy responses
Political	Fragmented coordination among agencies; limited local authority	Establish an inter-agency rural electrification taskforce; empower provincial governments to expedite project approvals.
Economic	Insufficient funding for remote villages; high infrastructure costs	Introduce blended finance models combining public, private, and donor funds; promote decentralized renewable systems to reduce grid extension costs.
Social	Exclusion of Hill Tribe communities from decision-making; lack of awareness	Implement participatory planning processes; provide community energy education and capacity-building programs.
Technological	Difficult terrain and grid extension challenges	Encourage deployment of off-grid and hybrid microgrid systems adapted to mountainous topography.
Environmental	Electrification restrictions in protected forest zones	Develop "green electrification" guidelines balancing conservation with essential infrastructure; use low-impact renewable technologies.
Legal	Complex permitting procedures; unclear land tenure for Hill Tribe settlements	Simplify and decentralize approval processes; reform land-use policies to allow basic infrastructure on community lands with environmental safeguards.

uncertainties with implementation. The results indicate that achieving 100% electrification in MHS is fundamentally a governance and policy coordination challenge, rather than solely an engineering or financing problem.

The study's contribution to sustainable electrification research

This study offers several distinct contributions to the literature on sustainable rural electrification. First, it provides one of the few systematic, empirically grounded analyses of electrification barriers in a legally complex and environmentally protected province in Thailand. Second, by applying the PESTEL framework to rural electrification, the study demonstrates the value of integrated, multi-dimensional analytical approaches for understanding energy access challenges in sensitive regions. Third, the explicit focus on institutional, legal, and land-use constraints advances current research, which often prioritizes technical feasibility while under examining governance dynamics.

The study's qualitative design enables the incorporation of expert knowledge and contextual nuance that are frequently absent from large-scale quantitative assessments. As such, the findings offer transferable insights for other mountainous and conservation-regulated regions in Southeast Asia and comparable settings globally.

Policy and implementation implications

The findings suggest that accelerating rural electrification in MHS Province requires regulatory flexibility, institutional coordination, and targeted financial mechanisms. Although the Provincial Energy Action Plan outlines pathways for electrifying remaining villages, current funding levels remain insufficient to meet identified needs. Streamlining approval procedures for small-scale renewable energy systems—particularly solar-based microgrids—could substantially reduce implementation delays while preserving environmental safeguards.

Enhanced coordination among the MOE, PEA, PAO, and local communities is essential to reduce administrative fragmentation. The establishment of dedicated funding instruments for off-grid and hybrid renewable systems, supported by national and international partners, could further address financing gaps. Collectively, these measures would support Thailand's commitments to SDG 7, while also contributing to SDGs 1, 2, 3, 4, 6, 10, and 13 by promoting inclusive, resilient, and sustainable rural development.

Limitations of the study

This study has several limitations that should be considered when interpreting the results. First, the inability to conduct site visits in three southern districts constrained direct observation of local conditions. However, interviews conducted with officials and residents in northern and central districts suggest that the identified barriers are broadly representative of provincial conditions. Second, although purposive sampling ensured institutional diversity, reliance on availability and official referrals may have limited representation of informal or marginalized community perspectives. Third, the absence of household-level survey data restricted direct quantitative assessment of socio-economic impacts. These limitations point to opportunities for methodological triangulation in future research.

Recommendations and future research directions

Future research should focus on developing and testing adaptive policy instruments that directly address the institutional and spatial constraints identified in this study. Priority areas include reforming land-use regulations to enable renewable microgrid deployment in protected areas, designing incentive mechanisms for community participation, and systematically evaluating the socio-economic outcomes of decentralized electrification.

Comparative studies across provinces with similar ecological and regulatory conditions would enhance generalizability and support national energy planning. Additionally, integrating quantitative approaches, such as techno-economic optimization models, lifecycle cost-benefit analyses, and impact evaluations, would strengthen the empirical foundation for investment and policy decisions. Overall, the findings emphasize that sustainable rural electrification depends not only on technological solutions, but also on governance reform, stakeholder engagement, and coherent cross-sectoral policy alignment.

Abbreviations

DEDE	Department of Alternative Energy Development and Efficiency
DNP	Department of National Parks, Wildlife and Plant Conservation
EGAT	Electricity Generating Authority of Thailand
EIA	Environmental Impact Assessment
ESMAP	Energy Sector Management Assistance Program
GPS	Global Positioning System
IRENA	International Renewable Energy Agency
LAO	Local Administrative Organization
LCOE	Levelized Cost of Electricity
MHS	Mae Hong Son
MOE	Ministry of Energy
NGO	Non-Governmental Organization
PAO	Provincial Administrative Organization
PEA	Provincial Electricity Authority
PESTEL	Political, Economic, Social, Technological, Environmental, Legal
PV	Photovoltaic
RET	Renewable energy technology
RFD	Royal Forest Department
SDG	Sustainable Development Goal
SHS	Solar home system

TIC Total installed cost
UNDP United Nations Development Programme

CRediT authorship contribution statement

Paul G. Marshall: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Prapita Thanarak:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Nipon Ketjoy:** Validation, Supervision, Resources, Investigation. **Unchittha Prasatsap:** Resources, Investigation.

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